# CSCI 580 - Artificial Intelligence
# Assignment #3
# Simulated Annealing (SA) + Genetic Algorithm (GA)

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## Overview

This assignment solves the same TSP instance using two heuristic optimization methods and implemented:

1. **Simulated Annealing (SA)** with a 2-opt neighborhood.
2. **Genetic Algorithm (GA)** on permutation tours (a standard evolutionary approach for TSP).
3. **Nearest Neighbor (NN)**, which is a greedy baseline heuristic.

## Implementation

The **TSP** instance generated and used 40, 2D cities with coordinates in $[0, 1] \times [0, 1]$, with a fixed seed in order to be reproduced. Here, a nearest neighbor greedy heuristic provided the baseline reference point tour with a length of approximately 6.0214.

The **Simulated Annealing (SA)**, begins from the nearest neighbor tour and creates new candidates by reversing random segments (2-opt swap). Here, the improving moves are always accepted, unlike the worse moves which are accepted with probability $exp(-\Delta/T)$. The temperature begins at 0.2 and degrades and breaks down by a factor of 0.9995 with each iteration, with the algorithm running for over $20,000$ times.

The **Genetic Algorithm (GA)**, utilizes a population of 200 tours, seeded with the nearest neighbor tour and filled with random variations. Here, each generation, the top 5 individuals are kept because of their superiority. Then, new off springs are then created using tournament selection ($k = 5$), with a cross over rate of 90%, and a swap mutation rate of 20%. Additionally, the algorithm is ran for over 400 generations.

# Results

Showing the output that is produced and presented in the terminal. The nearest neighbor baseline had a tour length value of 6.0214. The Simulated Annealing improved a lot, getting a best tour length of 5.0057, which is a 16.87% improvement. On the other hand, the Genetic Algorithm achieved a best tour length of 5.9001, with a smaller improvement of 2.01%.

That being said, SA used $20,000$ overall function evaluations while GA used over $80,200$. Meaning that SA found a better solution with a lot fewer total/overall evaluation. And we can see the following results here:

```
● @jldbautista → /workspaces/CSCI580_Joseph_Bautista/Assignment_4 (main) $ python assignment_4.py
  Nearest-Neighbor Length: 6.0214
  ============================================================
  Running Instrumented Simulated Annealing
  ============================================================
  [SA] iter=  2000  T=0.07356  cur=8.3800  best=6.0214
  [SA] iter=  4000  T=0.02705  cur=6.5972  best=6.0214
  [SA] iter=  6000  T=0.00995  cur=5.3139  best=5.3139
  [SA] iter=  8000  T=0.003659  cur=5.0057  best=5.0057
  [SA] iter= 10000  T=0.001346  cur=5.0057  best=5.0057
  [SA] iter= 12000  T=0.000495  cur=5.0057  best=5.0057
  [SA] iter= 14000  T=0.0001821  cur=5.0057  best=5.0057
  [SA] iter= 16000  T=6.696e-05  cur=5.0057  best=5.0057
  [SA] iter= 18000  T=2.463e-05  cur=5.0057  best=5.0057
  [SA] iter= 20000  T=9.057e-06  cur=5.0057  best=5.0057

  SA Best length: 5.0057
  SA Improvement vs NN: 16.87%

  ============================================================
  Running Instrumented Genetic Algorithm
  ============================================================
  [GA] gen=  50 best=5.9209 avg=6.1771 std=0.6236
  [GA] gen= 100 best=5.9001 avg=6.2211 std=0.7783
  [GA] gen= 150 best=5.9001 avg=6.1876 std=0.7326
  [GA] gen= 200 best=5.9001 avg=6.1951 std=0.7516
  [GA] gen= 250 best=5.9001 avg=6.1957 std=0.6842
  [GA] gen= 300 best=5.9001 avg=6.1961 std=0.6970
  [GA] gen= 350 best=5.9001 avg=6.1004 std=0.5631
  [GA] gen= 400 best=5.9001 avg=6.1935 std=0.7008

  GA Best Length: 5.9001
  GA Improvement vs NN: 2.01%

  Total Evaluations - SA: 20000 and GA: 80200

  ============================================================
  Grade Check
  ============================================================
  SA Length: 5.0057 | Improvement vs NN: 16.87%
  ✅ SA passed improvement threshold!
  GA Length: 5.9001 | Improvement vs NN: 2.01%
  ✅ GA passed improvement threshold!

  ============================================================
  Summary
  ============================================================
  NN: 6.0214 | SA: 5.0057 | GA: 5.9001
```

**Figure 1: SA Delta Histogram**

In figure 1, it shows the distribution of $\Delta E$ values that for accepted and rejected moves. In the accepted histogram, we can see that it has a large spike of negative deltas, meaning that most of the accepted moves were improvements.

But with that, we can also see that there's a visible spread of positive deltas that were accepted, meaning that SA did perform and take uphill moves to explore the search space.

Now for the rejected histogram, we can see that it only contains positive deltas, which were moves that worsened and the algorithm choosing not to accept. And we can see both of the histograms below.

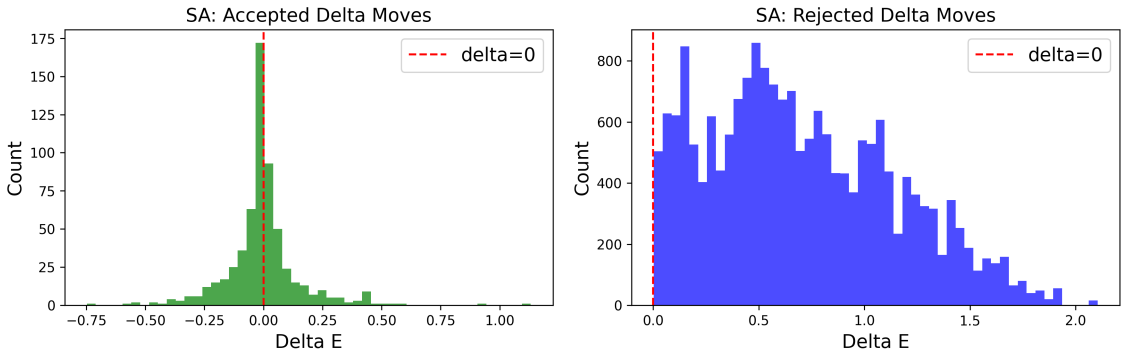



**Figure 2: SA Current vs Best Solution Over Time**

In figure 2, it shows the current solution (in orange) against the best that was found so far (in blue). In the first $6,000$ iterations, the current solution shifts and varies wildly, with shifts as high as 8+ while the best maintained 6.02.

This type of result shows that SA is accepting worse solutions at high temperatures in order to explore. With that, around iteration $6000 - 8000$, the current solution goes down and settles down and the best to its final value of 5.0057. And after that, both lines gradually stay flat as the temperature is far too low to accept any uphill moves. And we can see that below.

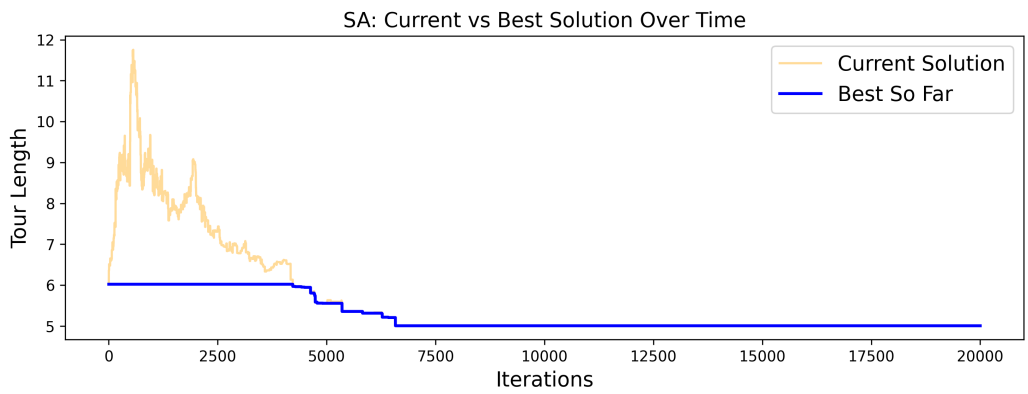



**Figure 3: SA Temperature Schedule**

In figure 3, it shows the geometric decay and degrade of the temperature from 0.2 towards 0. Here we can see that the curve drops highly at first but then flattens out as it reaches 0. This type of movement matches the behavior in Figure 2, when the temperature is high, SA explores with freedom, and as it cools down, the algorithm becomes more and more greedy and will only accept improvements. And we can see that below.

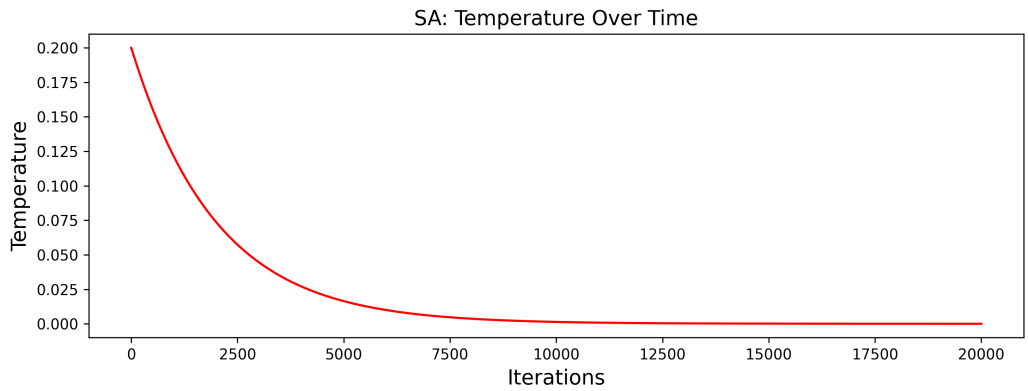



**Figure 4: GA Population Average Fitness Over Generations**

In figure 4, it follows the best (in blue), the average (in orange), and the worst (in red) tour lengths across generations. Here the best fitness drops a bit in the first 100 generations from $6.02 - 5.90$, but then stays completely flat. On the other hand, the average stays around $6.1 - 6.2$ and the worst staying around $8.0 - 9.0$ throughout the runs.

We can see that the gap between the best and the average remains somewhat constant, showing that the population never really converged fully or tightly but also never improved significantly. And we can see the figure below.

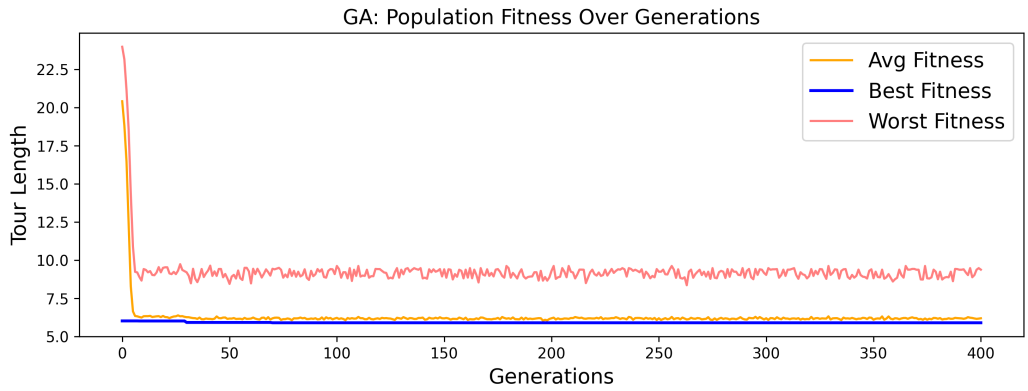



**Figure 5: GA Population Diversity**

In figure 5, it plots the standard deviation of tour lengths in the population over generations. We can see that the value shifts and fluctuates in the whole run but doesn't really show a clear downward trend, with it staying around $0.5 - 0.8$. And with Figure 4, this shows that while the population is maintained and kept, some spread in the fitness values, and it was not making good enough off springs to push the best solution lower. And we can see the figure below.

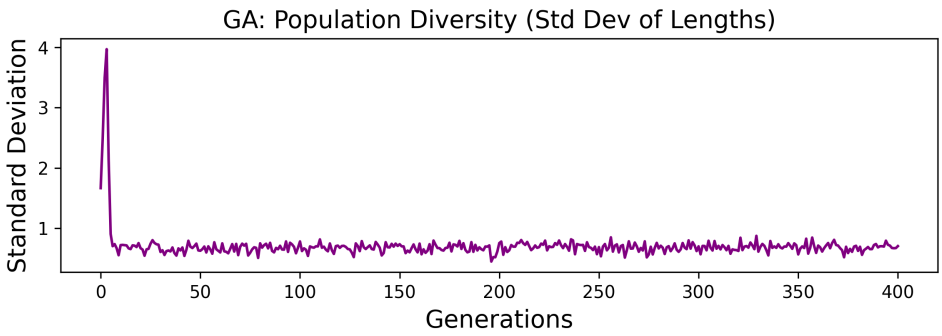



**Figure 6: SA vs GA - Normalized Comparison**

In figure 6, it overlays and compares both of the algorithms on the same $x - axis$ normalized from 0 to 1. Here we can see that the SA's curve begins at 6.02 and drops pretty steeply around the $25 - 40\%$ mark before flattening out at 5.0057.

On the other hand, GA's curve begins at 6.02 but only drops a bit to 5.9001, really early on, and then stays flat for the remaining of the run.

With the figure and result, we can see that SA found a much better solution and did it a lot more efficiently. And we can see the figure below.

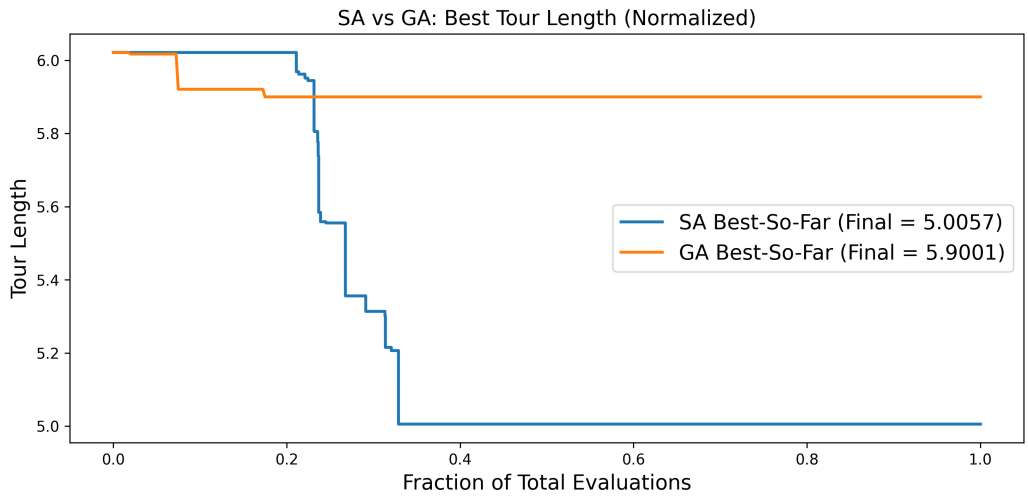
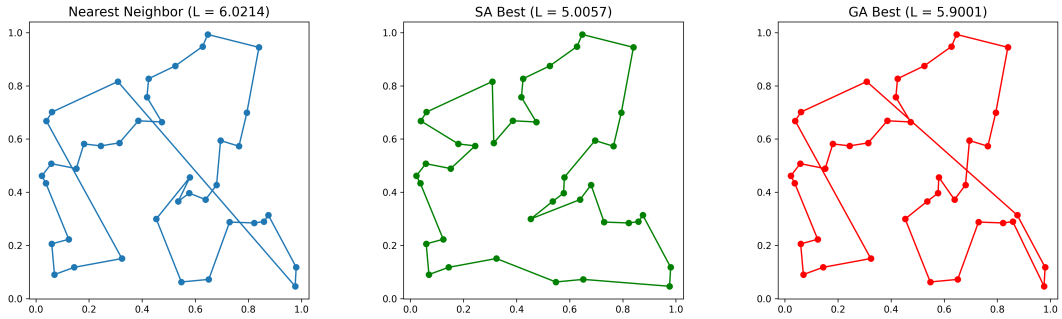



**Figure 7: Final Tour Plot Side by Side**

In figure 7, it shows the actual routes for all 3 of the different methods side by side. We can see that the NN tour (left) has some visible edge crossings due to its greedy structure.

With that, we can see that the SA tour (center), is much cleaner with fewer crossings, which is reflected by its length result of 5.0057.

And at last, the GA tour (right) looks the same to the NN tour but with smaller improvements, and is pretty consistent with its gain of 2.01%. And we can see the following figure below.




# Question Analysis

1. Which algorithm converged faster?

   The Simulated Annealing (SA) algorithm converged faster. It found the best solution of 5.0057 in only 8000 iterations, and didn't use up its total iteration amount of $20,000$. In comparison to the Genetic Algorithm (GA), using $80,200$ evaluations and only getting 5.9001.

2. Which showed more variability?

   The Simulated Annealing (SA) showed more variability. As the temperature cooled down, the variability decreased and the current solution balanced and stabilized.

3. Did SA escape local minima? Evidence?

   Yes, the SA escaped the local minima. We can see that in Figure 2, the SA frequently accepts worse solutions in the early iterations and runs, with the current solution going well above the best. This shows that SA moved away from the good solutions in order to explore new areas. And we can see more evidence in Figure 1, with the accepted delta histogram showing many positive deltas being accepted, further confirming that the SA took uphill moves.

4. Did GA lose diversity? Evidence?

   Yes, the GA lose diversity. We can see that in Figure 5, as it shows the population diversity shifting and fluctuating but not really leading towards improvements, with the standard deviation being around $0.5 - 0.8$ without being able to make better off springs. And with that, more evidence can be seen in Figure 4, as the average fitness stayed around $6.1 - 6.2$, while the best as stuck at 5.9001 for over 300 generations. This means that the population couldn't make enough diverse solutions through crossover and mutation in order to get out of its plateau.